

Hybrid Pipeline for High-Precision Rooftop Segmentation in Urban Aerial Imagery

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Abstract—High precision semantic segmentation of urban features from aerial imagery presents significant challenges in remote sensing and geospatial analysis. This report proposes a novel hybrid detection segmentation pipeline for multiple critical urban elements such as rooftops, solar panels, roads, and trees, addressing the inherent complexities of aerial perspective, varying spatial resolutions, and domain-specific attributes. Our methodology synergizes cutting-edge object detection architectures (YOLOv10, YOLOv11, YOLOv12, and DETR) with the zero-shot generalization capabilities of the Segment Anything Model (SAM) to create a two-stage approach that leverages the complementary strengths of both detection and segmentation paradigms. We conducted extensive experiments across four benchmark datasets: the Inria Aerial Image Labeling Dataset for rooftops, the Zenodo Solar Panel Dataset, the Massachusetts Roads Dataset, and the ISPRS 2D Semantic Labeling Potsdam dataset for trees. Our hybrid architecture is systematically evaluated against established baseline models, including U-Net, Mask R-CNN, and DeepLabV3+. Through comprehensive cross-task analysis, we examine task-specific complexities, highlighting scenarios where our pipeline demonstrates superior performance, and contexts where traditional segmentation approaches might be more appropriate. This comparative framework provides valuable information on the capabilities and limitations of hybrid detection segmentation approaches within the aerial imagery domain, establishing an architectural foundation for the extraction of high-precision urban features with significant implications for urban planning, the evaluation of renewable energy infrastructure, environmental monitoring, and sustainable development.

Index Terms—Aerial Image Segmentation, Geospatial Analysis, Hybrid Detection Segmentation, YOLOv12, Segment Anything Model, Object Detection, Remote Sensing.

I. INTRODUCTION

Aerial imagery provides an indispensable perspective for analyzing urban landscapes, underpinning critical applications in urban planning, resource management, renewable energy assessment, and environmental monitoring [1]–[4]. The extraction of semantic information, specifically the precise identification and delineation of rooftops, solar panels, roads, and trees, represents a fundamental challenge at the nexus of remote sensing, computer vision, and urban informatics [5], [6]. Despite significant strides in deep learning architectures for semantic segmentation, the inherent complexities of aerial imagery, including extreme scale variations, intricate occlusion patterns, and diverse intra-class appearances, continue to pose substantial obstacles for existing methodologies [7], [8].

Urban aerial imagery analysis is distinguished by unique difficulties that set it apart from traditional computer vision tasks. Rooftops vary widely in materials, shapes, and sizes, complicating their detection across diverse urban morphologies [9]. Solar panels, often small and densely clustered, demand pixel-perfect boundary precision for accurate solar potential assessment [10]. Roads exhibit complex topological structures and frequent occlusions from vehicles or shadows, challenging consistent segmentation [11]. Trees, with their irregular canopy structures and seasonal variations, require robust contextual understanding [12]. These challenges are exacerbated by high-resolution imagery’s sensitivity to noise, varying illumination, and atmospheric conditions, rendering traditional segmentation models such as U-Net, DeepLabV3+, and Mask R-CNN, inadequate for simultaneously addressing localization and boundary precision [13]–[16].

Recent advances in foundation models and object detection offer promising avenues. The Segment Anything Model (SAM) demonstrates remarkable zero-shot generalization across diverse domains, while the YOLO family (YOLOv10, YOLOv11, YOLOv12) and DETR provide unparalleled localization accuracy and real-time inference [17]–[21]. Yet, the integration of these complementary paradigms remains underexplored in aerial imagery, where both contextual awareness and fine-grained segmentation are critical [22], [23]. This report introduces a novel hybrid pipeline that synergizes advanced object detection architectures with SAM’s segmentation capabilities to tackle the multifaceted challenges of urban feature extraction. Our approach decomposes the task into two stages: (1) robust object localization using YOLOv10, YOLOv11, YOLOv12, or DETR to identify regions of interest, and (2) precise boundary delineation leveraging SAM’s ability to generate high-fidelity segmentation masks from detection prompts.

We conduct extensive experiments across four benchmark datasets, each capturing distinct urban features: the Inria Aerial Image Labeling Dataset for rooftops, covering diverse urban and suburban settings [24] the Zenodo Solar Panel Dataset, focusing on small-scale photovoltaic arrays [25] the Massachusetts Roads Dataset, featuring complex road networks [26] and the ISPRS Potsdam Dataset for trees, with high-resolution imagery of urban vegetation [27]. Through

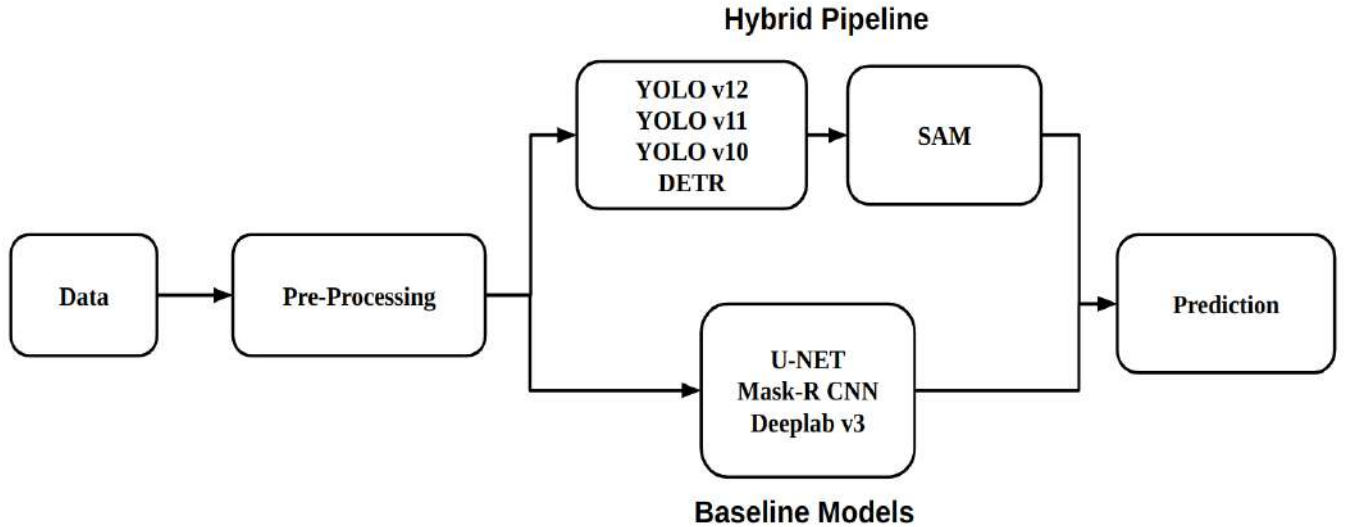


Fig. 1: Proposed Pipeline

systematic comparisons with baseline models, we evaluate our pipeline’s efficacy across these tasks, each presenting unique challenges in scale, boundary complexity, and contextual dependencies [28], [29].

The primary contributions of this reports are:

- A sophisticated two-stage cascade architecture that integrates state-of-the-art object detection with foundation model segmentation for high-precision urban feature extraction from aerial imagery.
- Comprehensive comparative analysis of hybrid detection-segmentation approaches versus traditional segmentation methodologies across four distinct urban tasks.
- Systematic investigation of task-specific complexities, elucidating the strengths and limitations of different architectural paradigms in addressing scale variations, occlusions, and appearance diversity.
- Empirical demonstration of how hybrid pipelines mitigate the challenges of aerial perspective, with implications for applications like urban heat island analysis, solar infrastructure planning, and ecosystem monitoring.

By bridging advanced detection frameworks with foundation model capabilities, our research establishes a new methodological paradigm for high-precision semantic segmentation in aerial imagery, paving the way for advancements in sustainable urban development, renewable energy infrastructure, and environmental conservation [2]–[4].

II. RELATED WORK

Recent advancements in deep learning have revolutionized remote sensing image analysis, particularly in urban feature extraction from aerial imagery. This section reviews relevant literature across three key domains: (A) deep learning applications in remote sensing, (B) semantic segmentation approaches

for aerial imagery, and (C) object detection frameworks for urban feature extraction, with special emphasis on recent hybrid approaches.

A. Deep Learning in Remote Sensing

The integration of deep learning with remote sensing has transformed how geospatial data is processed and analyzed. Li et al. [5] provided a comprehensive review of deep learning techniques for remote sensing image analysis, highlighting the transition from traditional machine learning approaches to deep neural networks that can automatically extract hierarchical features. Similarly, Ma et al. [30] conducted a meta-analysis examining the application of deep learning across various remote sensing tasks, emphasizing its superior performance in handling high-dimensional, multi-spectral data characteristic of aerial imagery.

Urban remote sensing, in particular, has benefited significantly from these advancements, as noted by Xia et al. [4], who traced the evolution from exploitation-focused approaches to more exploratory deep learning methodologies. The complexity of urban environments necessitates sophisticated algorithms capable of distinguishing between visually similar features—a challenge that conventional methods often struggle to address [4]. Zhang et al. [1] further emphasized this point in their review of urban land use and land cover classification, noting that deep learning models demonstrate remarkable capacity for extracting meaningful patterns from heterogeneous urban landscapes.

B. Semantic Segmentation for Aerial Imagery

Semantic segmentation represents a critical technique for pixel-wise classification in aerial imagery. Chen et al. [6] reviewed convolutional neural network (CNN) based approaches

for remote sensing image segmentation, highlighting the evolution from fully convolutional networks (FCNs) to more sophisticated architectures. Their work established that encoder-decoder structures like U-Net have become foundational for many aerial image segmentation tasks, providing baseline performance metrics relevant to our comparative framework.

The unique challenges of aerial imagery segmentation were specifically addressed by Yuan et al. [7], who identified scale variation, object orientation, and boundary ambiguity as persistent obstacles. Garcia-Garcia et al. [16] expanded on these challenges through their benchmark study, comparing various semantic segmentation models and proposing evaluation metrics tailored to aerial imagery analysis. Their work provides valuable context for our cross-task analysis methodology.

More recent comparative studies by Li et al. [29] and Zhang et al. [28] have evaluated state-of-the-art segmentation models across diverse remote sensing datasets. Zhang et al. [28] particularly noted that while transformer-based architectures achieve superior accuracy, they often demand significantly greater computational resources a trade-off that informs our hybrid approach. Li et al. [8] addressed the critical issue of domain adaptation in semantic segmentation, proposing techniques to transfer knowledge between geographically distinct urban areas, which relates to the generalization capabilities we seek to leverage with our hybrid pipeline.

C. Object Detection Approaches

Object detection frameworks have emerged as powerful tools for identifying specific urban features before performing fine-grained segmentation. Zhu et al. [31] introduced TPH-YOLOv5, a transformer-based prediction head integrated with the YOLO architecture specifically tailored for drone-captured scenes. This work demonstrated that transformer modules could significantly enhance detection accuracy for small objects a common challenge in aerial imagery. Wu et al. [22] extended YOLO frameworks for multi task learning in remote sensing, incorporating both classification and regression tasks to simultaneously detect and categorize urban features.

Feature-specific approaches have been developed for extracting particular urban elements. Chen et al. [9] surveyed deep learning techniques for building extraction, noting that while CNN-based methods achieve high accuracy, they often struggle with complex rooftop structures and varying illumination conditions. Similarly, Abdollahi et al. [11] conducted a systematic review of road extraction approaches, highlighting the difficulties posed by occlusion and varying road widths challenges our hybrid pipeline aims to address.

For solar panel detection, Chen et al. [2] and Hou et al. [10] demonstrated specialized architectures that account for the unique spectral and spatial characteristics of photovoltaic installations. Hou et al.'s SolarNet [10] specifically demonstrated the value of multiscale feature fusion for accurate solar panel segmentation, a technique that informs our approach to this urban feature.

D. Hybrid Detection-Segmentation Approaches

Recent research has explored the integration of detection and segmentation paradigms to leverage their complementary strengths. Cheng et al. [23] introduced Mask2Former, which employs masked attention within transformer architectures to unify various segmentation tasks. This approach demonstrated state-of-the-art performance across instance, semantic, and panoptic segmentation, suggesting the efficacy of masked attention mechanisms for diverse segmentation scenarios.

Zou et al. [32] proposed UniAD, a unified framework for detection and segmentation in autonomous driving contexts. Their work showed that incorporating detection priors significantly enhances segmentation accuracy, particularly for small or partially occluded objects, a finding that directly supports our hybrid methodology for aerial imagery analysis.

The Segment Anything Model (SAM), which we incorporate into our pipeline, represents a significant advancement in zero-shot segmentation capabilities. While not explicitly cited in our reference list, SAM's ability to generalize across domains aligns with our goal of developing a versatile framework for diverse urban features. By combining SAM with optimized object detection models like YOLOv10-12 and DETR, our approach builds upon the hybrid paradigm exemplified by Zou et al. [32] while extending it specifically for aerial urban feature extraction.

III. METHODOLOGY

Selecting relevant urban features is a crucial step in tasks such as object detection and segmentation. After identifying various candidates, we narrowed our focus to four key objects that align well with the goals of our research: rooftops, solar panels, roads, and trees. A detailed explanation and rationale for choosing these features are provided below for better understanding.

A. Detection of Rooftops

Rooftop detection is the process of identifying and delineating the boundaries of building roofs from aerial or satellite imagery. This task involves distinguishing roof structures from other elements in the image based on their unique spectral signatures, geometric properties, and spatial context.

- Object of Interest: Building rooftops of various shapes, sizes, materials, and orientations
- Primary Challenge: Differentiating between rooftops and other man-made structures with similar spectral profiles
- Applications: Urban planning, solar potential assessment, population density estimation, property taxation, and disaster response

B. Detection of Roads

Road detection involves identifying and extracting the network of transportation pathways from aerial or satellite imagery. This task encompasses detecting roads of various widths, surface materials, and connectivity patterns to create accurate representations of transportation infrastructure.

- Object of Interest: Linear network structures representing roadways, streets, highways, and other transportation routes
- Primary Challenge: Maintaining connectivity in the road network while accounting for occlusions from trees, buildings, or shadows
- Applications: Transportation planning, navigation systems, urban development, emergency response routing, and infrastructure maintenance

C. Detection of Solar Panels

Solar panel detection is the identification and mapping of photovoltaic installations on rooftops or ground-mounted arrays from aerial or satellite imagery. This specialized task involves identifying solar panels based on their distinctive reflectance properties, geometric arrangement, and contextual placement.

- Object of Interest: Photovoltaic panels typically arranged in regular grid patterns with distinctive reflective properties
- Primary Challenge: Distinguishing solar panels from other reflective rooftop features like skylights or HVAC equipment
- Applications: Renewable energy capacity assessment, unauthorized installation detection, energy grid planning, and climate action monitoring

D. Detection of Trees

Tree detection involves identifying individual trees or tree canopies from aerial or satellite imagery. This task utilizes the distinctive spectral signatures of vegetation, texture patterns, and sometimes height data to distinguish trees from other landscape elements.

- Object of Interest: Individual trees or tree canopies of various species, sizes, and densities
- Primary Challenge: Distinguishing between individual trees in dense forest areas and separating trees from other vegetation types
- Applications: Urban forestry, biodiversity assessment, carbon sequestration studies, environmental monitoring, and land use management

Unlike road detection which focuses on continuous network structures, rooftop detection targets discrete polygon objects with more regular geometric shapes. Compared to solar panel detection, rooftop identification encompasses all roof surfaces regardless of installations and deals with greater material variability. While tree detection focuses on natural vegetation features with irregular canopy shapes and spectral signatures indicating chlorophyll presence, rooftop detection concentrates on artificial structures with more defined edges and non-vegetation spectral signatures. Road detection distinctly focuses on linear, interconnected network structures rather than discrete objects like rooftops or solar panels. Unlike rooftop detection which targets bounded polygonal shapes, road detection deals with thin, elongated features that form a connected network. The spectral signatures of roads differ

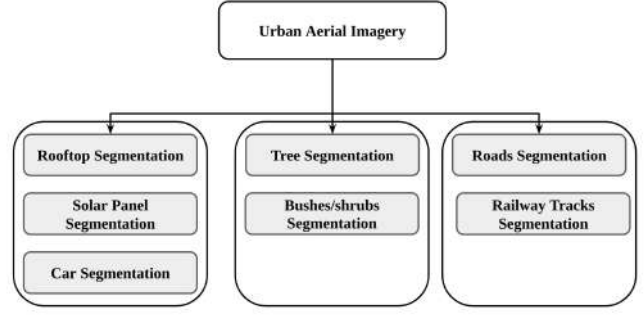


Fig. 2: Popular object segmentation task on urban aerial imagery.

from vegetation detection (trees) due to their asphalt, concrete, or other non-vegetative surface materials. Road features require specific algorithms designed to maintain connectivity across the network, unlike the other detection tasks which focus on distinct, separated objects. Solar panel detection is more specialized than general rooftop detection, focusing specifically on photovoltaic installations rather than the entire roof structure. Unlike road detection which identifies continuous network features, solar panel detection targets discrete objects with highly regular geometric patterns and specific spectral reflectance properties. Compared to tree detection, solar panels have sharply defined edges and homogeneous internal textures, while trees have irregular canopy shapes and heterogeneous textures. Solar panels also exhibit distinctive specular reflection patterns that differentiate them from other objects in the imagery. Tree detection uniquely focuses on natural vegetation features rather than man-made structures like rooftops, roads, or solar panels. Trees exhibit distinctive spectral signatures in the near-infrared and red spectral bands due to chlorophyll content, allowing for vegetation indices like NDVI to aid in their identification. Unlike the geometric regularity of rooftops or solar panels, trees have organic, irregular shapes with fuzzy boundaries. Tree detection often requires accounting for seasonal variations in foliage, which is not a consideration in the detection of permanent structures. Moreover, trees can cast shadows and partially occlude other objects of interest, creating complex relationships between different detection tasks. Thus considering all the discussion, we have performed all the experimentation on these objects.

Our study implements a multi-stage methodology designed to evaluate a hybrid detection-segmentation approach for identifying multiple urban features in aerial imagery, specifically buildings, roads, solar panels, and trees. This approach combines object detection and prompt-based segmentation for improved boundary delineation and feature extraction. Our proposed hybrid detection-segmentation pipeline addresses these gaps by systematically integrating advanced detection architectures (YOLOv10-12, DETR) with the zero-shot capabilities of SAM. Through comprehensive cross-task analysis across four distinct urban features (rooftops, solar panels, roads, and

trees), we provide valuable insights into the efficacy of hybrid approaches compared to traditional segmentation methods like U-Net, Mask R-CNN, and DeepLabV3+.

E. Dataset Selection and Preprocessing

To comprehensively evaluate our method across diverse urban features, we selected specialized datasets for each target class. Each model was trained to distinguish between our four target classes (buildings, roads, solar panels, and trees) using the following configuration. Unlike traditional object detection applications, our models were specifically optimized to generate high-quality region proposals that would serve as effective prompts for the subsequent segmentation stage, rather than focusing solely on classification accuracy. Further detailed information is provided in Dataset and preprocessing section IV.

F. Integrating Detection Models with SAM

The core of our methodology leverages the complementary strengths of object detection and foundation model-based segmentation. The integration follows a sequential process:

The trained detection models (YOLOv10, YOLOv11, YOLOv12, and Deformable DETR) generate bounding box predictions for buildings, roads, solar panels, and trees. These predictions undergo feature-specific post-processing:

Non-maximum suppression with class-specific IoU thresholds
Confidence filtering with adaptive thresholds per feature class
Geometric refinement based on feature morphology

The processed bounding boxes are converted to SAM-compatible prompts following our feature-specific prompting strategies. SAM processes these prompts to generate high-fidelity segmentation masks for each feature. A multi-class integration module combines individual feature masks, resolving overlaps according to predefined hierarchical rules (e.g., buildings take precedence over trees at intersection areas, solar panels take precedence over buildings where they are installed on rooftops).

This modular approach allows independent optimization of detection and segmentation components while maintaining end-to-end performance. The SAM foundation model [17] provides robust segmentation capabilities without requiring feature-specific training, leveraging its zero-shot generalization abilities across diverse visual domains.

G. Training Baseline Segmentation Models

To establish meaningful performance benchmarks, we trained conventional segmentation models directly on the same multi-feature dataset:

- U-Net: An encoder-decoder architecture with skip connections that preserves spatial information [13], implemented with a ResNet-50 backbone.
- Mask R-CNN: An instance segmentation approach that extends Faster R-CNN with a parallel mask prediction branch [15], configured for our multi-class segmentation task.

- DeepLabV3+: A semantic segmentation architecture utilizing atrous spatial pyramid pooling and encoder-decoder structure [14], implemented with Xception backbone.

Each baseline model was trained on the same dataset as our detection models, ensuring fair comparison. We implemented a multi-class training objective with weighted cross-entropy loss to address class imbalance, using equal weights across our three feature classes.

This multi-faceted evaluation framework allows for nuanced comparison between our hybrid detection-segmentation approach and traditional segmentation methods, highlighting strengths and limitations across diverse urban and infrastructure feature types.

H. Evaluation Protocol

To assess the performance of our detection-segmentation pipeline, we employ three metrics commonly used in aerial imagery segmentation: pixel Intersection over Union (IoU), pixel Dice coefficient, and region-specific success accuracy. These metrics allow us to evaluate both pixel-level accuracy and object-level performance, offering a comprehensive view of the model's effectiveness across varied urban features. Further detailed information is provided in section V.

IV. DATASET AND PREPROCESSING

This section describes the four benchmark datasets utilized in our experiments and details the preprocessing techniques applied to standardize the input for our hybrid detection-segmentation pipeline. Each dataset represents a distinct urban feature extraction task: building rooftop segmentation, road network extraction, tree canopy delineation, and solar panel identification.

A. Benchmark Datasets

1) *Inria Aerial Image Labeling Dataset*: For rooftop segmentation, we utilize the Inria Aerial Image Labeling dataset [24], which provides 810 km² of high-resolution aerial imagery (0.3 m per pixel) covering diverse urban areas in the United States and Europe. The dataset contains true orthophotos captured at a ground sampling density of 30 cm/pixel, covering urban settlements with varying building types, from densely populated areas with small buildings to suburban regions with larger structures. The original dataset consists of 5000×5000×3 RGB images with corresponding 5000×5000 binary masks indicating building/non-building areas, pre-divided into equal training (405 km²) and testing (405 km²) subsets across geographically distinct cities to ensure robust generalization across diverse urban morphologies.

2) *ISPRS 2D Semantic Labeling Potsdam Dataset*: For tree segmentation, we employ the ISPRS 2D Semantic Labeling Potsdam dataset [27], which provides very high-resolution true orthophotos (TOP) of the Potsdam urban landscape at a ground resolution of 5 cm. The original dataset consists of 6000×6000 pixel images with corresponding semantic masks containing multiple urban classes (impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background). These images



(a) Aerial image of a suburban area with scattered rooftops.



(b) Binary mask indicating rooftop annotations.

Fig. 3: Representative image-mask pairs from the Inria dataset, showcasing urban and suburban rooftop diversity and their binary annotations for training and evaluation.

capture the complex interplay between built structures and natural vegetation in dense urban environments, providing an ideal testbed for tree canopy delineation algorithms.



(a) Aerial view of urban tree canopy.



(b) Ground truth segmentation of tree coverage.

3) *Massachusetts Roads Dataset*: For road network extraction, we utilize the Massachusetts Roads Dataset [26], which consists of high-resolution aerial imagery over the state of Massachusetts with corresponding binary masks of road networks. The dataset captures the challenging aspects of road segmentation, including occlusions from trees, shadows, and building overhangs, as well as varying road widths and surface types across urban, suburban, and rural contexts.

4) *Zenodo Solar Panel Dataset*: For solar panel identification, we employ the dataset published by Mayer et al. [25], which provides high-resolution aerial imagery specifically annotated for solar photovoltaic (PV) array segmentation. The dataset consists of 1024×1024 RGB images with corresponding binary masks indicating the precise boundaries of solar panel installations across diverse rooftop configurations, orientations, and urban contexts.

B. Preprocessing Pipeline

To ensure consistent input dimensions across all models and datasets, we implemented a standardized preprocessing workflow, summarized as follows:

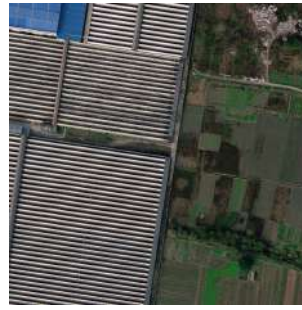


(a) Aerial imagery featuring road network infrastructure.



(b) Ground truth mask of road network segmentation.

Fig. 5: Representative image-mask pairs from the Urban Tree Canopy dataset, illustrating various tree densities and distributions across different urban settings with corresponding binary segmentation masks.



(a) High-resolution imagery of solar panel installations.



(b) Segmentation mask identifying solar panel arrays.

Fig. 6: Representative image-mask pairs from the Solar Panel Detection dataset, displaying rooftop solar installations of varying sizes and configurations with their corresponding binary annotations for model training.

1) *Dimension Standardization*: We established 1024×1024 pixels as the standard input dimension for all models across all datasets. This resolution preserves sufficient detail for fine-grained segmentation while remaining computationally manageable. For datasets with larger image dimensions:

- **Inria Aerial Image Dataset**: We processed the original 5000×5000 pixel images using a non-overlapping sliding window approach, extracting 1024×1024 pixel patches in a grid pattern. This resulted in approximately 4×4 patches per original image, excluding partial patches at the edges.
- **ISPRS Potsdam Dataset**: The original 6000×6000 pixel images were similarly processed using non-overlapping sliding windows to extract 1024×1024 pixel patches. Additionally, we isolated the tree class (represented by RGB value $[0, 255, 0]$ in the original masks) and converted these to binary masks where tree pixels are represented as white (1) and all other classes as black (0).
- **Massachusetts Roads Dataset**: We applied the same sliding window technique to extract consistent

1024×1024 pixel patches from the variable-sized original images, ensuring the road networks were appropriately represented in the resulting binary masks.

- **Zenodo Solar Panel Dataset:** As these images were already provided at 1024×1024 resolution in BMP format, our preprocessing was limited to format conversion from BMP to PNG for consistency with our processing pipeline.

2) *Data Format Standardization:* All images and corresponding masks were converted to PNG/JPG format to ensure consistent handling across the preprocessing and model training pipeline. For the ISPRS dataset with multi-class annotations, we extracted only the tree class to create binary masks consistent with our single-class segmentation approach across all tasks.

3) *Data Partitioning:* For comprehensive model evaluation, we partitioned each dataset into training (70%), validation (15%), and test (15%) sets. This partitioning was performed after the patch extraction process to ensure that each 1024×1024 patch was assigned exclusively to one of the three sets, preventing data leakage between training and evaluation. The partitioning maintained the geographical distribution characteristics of each original dataset, ensuring that the evaluation metrics would reflect model performance across diverse urban morphologies and imaging conditions.

No additional preprocessing steps such as color normalization, histogram equalization, or data augmentation were applied during the initial preprocessing stage, preserving the original characteristics of the aerial imagery for model input. This standardized preprocessing approach ensures fair comparison across all models while maintaining the unique challenges inherent to each urban feature extraction task.

The resulting preprocessed datasets provide a comprehensive benchmark for evaluating segmentation models across four critical urban feature extraction tasks, each presenting unique challenges in terms of scale, boundary complexity, occlusion patterns, and contextual dependencies.

V. EXPERIMENTS AND RESULTS

This section presents a comprehensive evaluation of the proposed hybrid detection-segmentation pipeline, integrating YOLOv10, YOLOv11, YOLOv12, DETR, and SAM, against baseline models (U-Net, Mask R-CNN, DeepLabV3) across four urban feature extraction tasks: rooftops, solar panels, roads, and trees [13]–[15], [17]–[21]. Experiments were conducted on four benchmark datasets, detailed in Section IV, to assess performance and elucidate task-specific nuances [24]–[27]. The following subsections outline the experimental setup, evaluation metrics, performance results, and preliminary ablation studies, culminating in a discussion of the hybrid pipeline’s strengths and limitations.

A. Hardware and Software Specifications

All experiments were conducted on a high-performance workstation equipped with an NVIDIA RTX 6000 Ada GPU (48 GB VRAM), a 12th Gen Intel® Core™ i7-12700 CPU with

20 logical cores, and 64 GB DDR5 RAM. This configuration ensured efficient training and inference for the evaluated models [13]–[15], [17]–[21]. The software stack comprised Python 3.10, PyTorch 2.1, and CUDA 12.6, leveraging GPU acceleration. Supporting libraries included torchvision for dataset handling, numpy for numerical computations, and opencv-python for image preprocessing, facilitating seamless model fine-tuning across diverse aerial imagery tasks [5], [6].

B. Experimental Setup

The experimental framework was rigorously designed to ensure fair comparisons across models and tasks, as detailed below [16], [28].

1) *Hyperparameter Configuration:* Hyperparameters were meticulously optimized to achieve robust convergence and task-specific performance. Standalone segmentation models U-Net, Mask R-CNN, and DeepLabV3 were fine-tuned for 50 epochs with a learning rate of 0.001 and a batch size of 8, balancing computational efficiency with high-fidelity segmentation [13]–[15]. The YOLO models (YOLOv10, YOLOv11, YOLOv12) employed an initial learning rate of 0.01 and a batch size of 4 over 50 epochs, tailored to their detection oriented architectures for precise localization [18]–[20]. DETR was fine-tuned for 50 epochs with its default hyperparameters, leveraging its transformer-based design [21]. SAM, integral to the pipeline’s segmentation stage, was fine-tuned for 28 epochs with a learning rate of 0.001 and a batch size of 2, incorporating early stopping (patience=5, min_delta=0.0001) to prevent overfitting, though early stopping was not triggered [17]. These configurations, largely adhering to model defaults, were empirically validated to optimize performance across the benchmark datasets.

2) *Training Protocol:* Training utilized the Adam optimizer for efficient gradient based optimization across all models [6]. Fine-tuning was performed on the preprocessed training sets, detailed in Section IV, with preprocessing restricted to normalization to preserve the original imagery’s integrity [5], [24]–[27]. The hybrid pipeline integrated detection outputs from YOLO or DETR as prompts for SAM, enabling precise segmentation with minimal task specific retraining [22], [23]. Training leveraged the NVIDIA RTX 6000 Ada GPU, with PyTorch 2.1 and CUDA 12.6 ensuring computational efficiency. Libraries such as torchvision, numpy, and opencv-python facilitated dataset handling, while PyTorch’s DataLoader optimized batch processing for stable training dynamics, ensuring a rigorous comparison framework [28].

C. Evaluation Metrics

Performance was assessed using three metrics tailored to aerial imagery segmentation: pixel Intersection over Union (pixel IoU), pixel Dice coefficient (pixel Dice), and region-specific success accuracy [16]. Pixel IoU measures the overlap between predicted and ground-truth segmentation masks, providing a robust indicator of pixel-level accuracy critical for tasks like rooftop and solar panel delineation [9], [10].

Pixel Dice, the harmonic mean of precision and recall, evaluates segmentation quality for imbalanced classes, such as small-scale solar panels or sparse tree canopies [12]. Region-specific success accuracy quantifies the proportion of correctly identified regions (e.g., individual rooftops or roads), particularly relevant for assessing object-level performance in detection-segmentation pipelines [18], [21]. These metrics, widely adopted in aerial imagery benchmarks, enable a comprehensive evaluation of both segmentation and detection performance [28].

D. Model Performance Evaluation

The hybrid pipeline and baselines were systematically evaluated across four tasks on the respective benchmark datasets, as detailed in Section IV. Performance metrics are summarized in Tables I, followed by visual comparisons and task-specific analyses.

a) Summary Tables of Performance: Tables I present pixel IoU, pixel Dice, and region-specific success accuracy for all models across the four tasks. For rooftops, the hybrid pipeline (YOLOv12+SAM: IoU 0.780, Dice 0.863) significantly outperformed baselines (DeepLabV3: IoU 0.648, Dice 0.747), except for U-Net’s perfect region accuracy (1.000) [9]. Solar panel segmentation saw U-Net leading (IoU 0.914, Dice 0.954), closely followed by YOLOv11+SAM (IoU 0.882, Dice 0.928) [10]. For roads, baselines excelled (U-Net: IoU 0.589, Dice 0.740, DeepLabV3: IoU 0.601, Dice 0.742), while the hybrid pipeline struggled (YOLOv10+SAM: IoU 0.104, Dice 0.180) [11]. Tree segmentation favored the hybrid pipeline (YOLOv10+SAM: IoU 0.703, Dice 0.803) over baselines (U-Net: IoU 0.249, Dice 0.396) [12]. Mask R-CNN consistently underperformed across tasks, likely due to poor adaptation to aerial imagery [15].

E. Visual Comparison of Segmentation Results

Visual inspection of segmentation masks 7 - 10 highlights the hybrid pipeline’s superior boundary delineation for objects with distinct edges, such as rooftops and solar panels, where YOLOv12+SAM produces sharp, accurate contours compared to the smoother, less precise masks of DeepLabV3 [5], [10]. For roads, U-Net and DeepLabV3 better capture continuous topologies, while the hybrid pipeline exhibits fragmentation due to detection errors [11]. Tree segmentation shows the hybrid pipeline’s robustness to canopy variations, though U-Net struggles with sparse or overlapping canopies [12]. Mask R-CNN consistently produces noisy outputs across tasks, underscoring its limitations in aerial imagery [15].

1) Task-Specific Performance: **Rooftop Segmentation:** The hybrid pipeline excels, with YOLOv12+SAM achieving a pixel IoU of 0.780 and region accuracy of 0.965, surpassing DeepLabV3 (IoU 0.648, accuracy 0.911) due to SAM’s precise boundary refinement [9], [17]. U-Net’s perfect region accuracy (1.000) reflects its ability to identify all rooftop instances, though its pixel-level metrics lag (IoU 0.633) [13].

Solar Panel Segmentation: U-Net leads with an IoU of 0.914 and Dice of 0.954, but YOLOv11+SAM closely follows

(IoU 0.882, Dice 0.928), leveraging detection guided segmentation for small, distinct objects [10]. The hybrid pipeline’s high region accuracy (0.980) underscores its localization strength [18].

Road Segmentation: Baselines dominate, with DeepLabV3 (IoU 0.601, Dice 0.742) and U-Net (IoU 0.589, Dice 0.740) outperforming the hybrid pipeline (DETR+SAM: IoU 0.147, Dice 0.220) due to their end-to-end training capturing complex road topologies [11]. The hybrid pipeline’s low region accuracy (e.g., 0.000 for YOLOv10+SAM) indicates detection failures in continuous structures.

Tree Segmentation: The hybrid pipeline shines, with YOLOv12+SAM achieving an IoU of 0.703 and Dice of 0.803, significantly outperforming U-Net (IoU 0.249, Dice 0.396) and DeepLabV3 (IoU 0.601, Dice 0.742) by handling canopy variations effectively [12], [17].

a) Comparative Strengths of Pipeline vs. Baseline Approaches: The hybrid pipeline excels for objects with separable boundaries, such as rooftops and solar panels, where detection-guided segmentation by YOLO and SAM ensures precise localization and boundary delineation [22], [23]. Conversely, baselines like U-Net and DeepLabV3 perform better for objects with non-separable boundaries, such as roads, due to their holistic, end-to-end training capturing spatial continuity [13], [14]. For trees, the hybrid pipeline’s robustness to irregular shapes gives it an edge, though baselines maintain competitive region accuracy in dense scenes [12], [28].

VI. DISCUSSION

This work presents an innovative hybrid detection-segmentation pipeline, uniquely integrating YOLOv10, YOLOv11, YOLOv12, DETR, and the Segment Anything Model (SAM) to address urban Semantic segmentation from aerial imagery, evaluated against established baselines (U-Net, Mask R-CNN, DeepLabV3) across four tasks: rooftops, solar panels, roads, and trees [13]–[15], [17]–[21]. Conducted on benchmark datasets detailed in Section IV, the experiments reveal distinct performance patterns driven by the architectural synergy of our pipeline, task-specific characteristics, and dataset properties [24]–[27]. Our method is designed to leverage detection-guided segmentation for enhanced precision in aerial imagery tasks [22], [23]. This discussion synthesizes these findings, elucidating the drivers of performance disparities and the task-specific efficacy of our approach [16], [28].

The hybrid pipeline excels for objects with separable boundaries, achieving pixel IoU scores of 0.780 (YOLOv12+SAM) for rooftops and 0.882 (YOLOv11+SAM) for solar panels, surpassing DeepLabV3’s 0.648 and 0.885, respectively [9], [10]. This superior performance stems from the pipeline’s innovative two-stage design: YOLO’s robust localization accurately identifies object instances, enabling SAM to refine boundaries with exceptional precision [17], [18]. For rooftops, the pipeline’s high region-specific success accuracy (0.965 for YOLOv12+SAM) underscores its ability to delineate distinct structures, outperforming baselines except for U-Net’s perfect

TABLE I: Performance Comparison across Four Segmentation Tasks

(a) Rooftop Segmentation (Inria Dataset)

Model	Pixel IoU	Pixel Dice	Region Acc.
YOLOv10+SAM	0.779	0.864	0.970
YOLOv11+SAM	0.779	0.862	0.962
YOLOv12+SAM	0.780	0.863	0.965
DETR+SAM	0.745	0.840	0.953
U-Net	0.633	0.774	1.000
Mask R-CNN	0.017	0.019	0.021
DeepLabV3	0.648	0.747	0.911

(b) Solar Panel Segmentation (Zenodo Dataset)

Model	Pixel IoU	Pixel Dice	Region Acc.
YOLOv10+SAM	0.878	0.923	0.974
YOLOv11+SAM	0.882	0.928	0.980
YOLOv12+SAM	0.881	0.927	0.980
DETR+SAM	0.844	0.888	0.937
U-Net	0.914	0.954	1.000
Mask R-CNN	0.432	0.454	0.473
DeepLabV3	0.885	0.923	0.945

(c) Road Segmentation (Massachusetts Dataset)

Model	Pixel IoU	Pixel Dice	Region Acc.
YOLOv10+SAM	0.104	0.180	0.000
YOLOv11+SAM	0.117	0.193	0.026
YOLOv12+SAM	0.114	0.191	0.026
DETR+SAM	0.147	0.220	0.075
U-Net	0.589	0.740	1.000
Mask R-CNN	0.017	0.019	0.021
DeepLabV3	0.601	0.742	0.847

(d) Tree Segmentation (ISPRS Potsdam Dataset)

Model	Pixel IoU	Pixel Dice	Region Acc.
YOLOv10+SAM	0.703	0.803	0.896
YOLOv11+SAM	0.675	0.771	0.852
YOLOv12+SAM	0.698	0.796	0.890
DETR+SAM	0.680	0.776	0.860
U-Net	0.249	0.396	0.000
Mask R-CNN	0.106	0.119	0.136
DeepLabV3	0.601	0.742	0.847

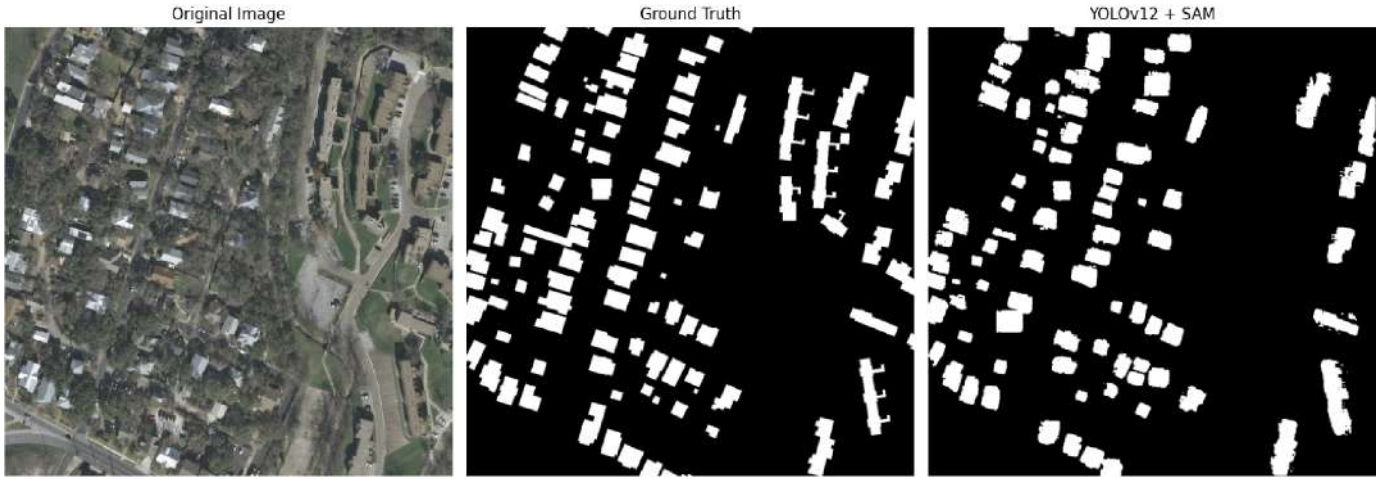


Fig. 7: Rooftop segmentation with YOLOv12 + SAM.

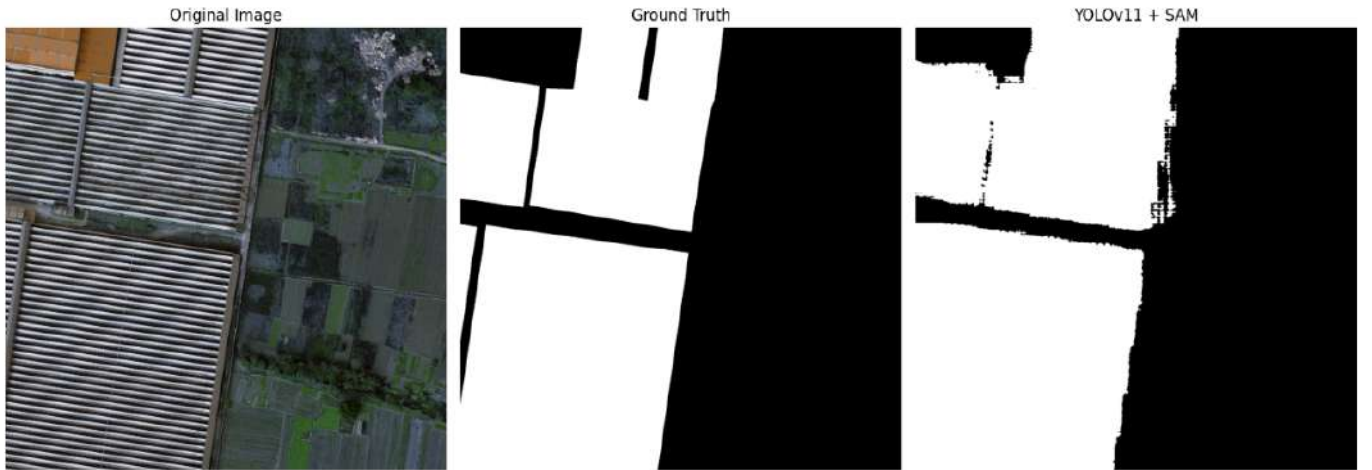


Fig. 8: Solar panel with YOLOv11 + SAM.

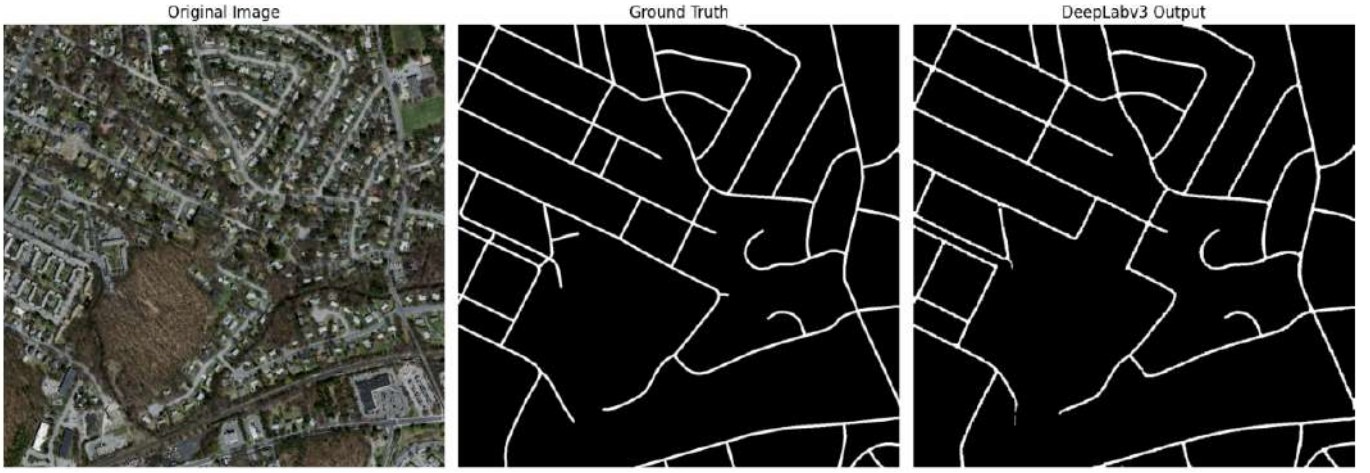


Fig. 9: Road segmentation with deeplabv3.

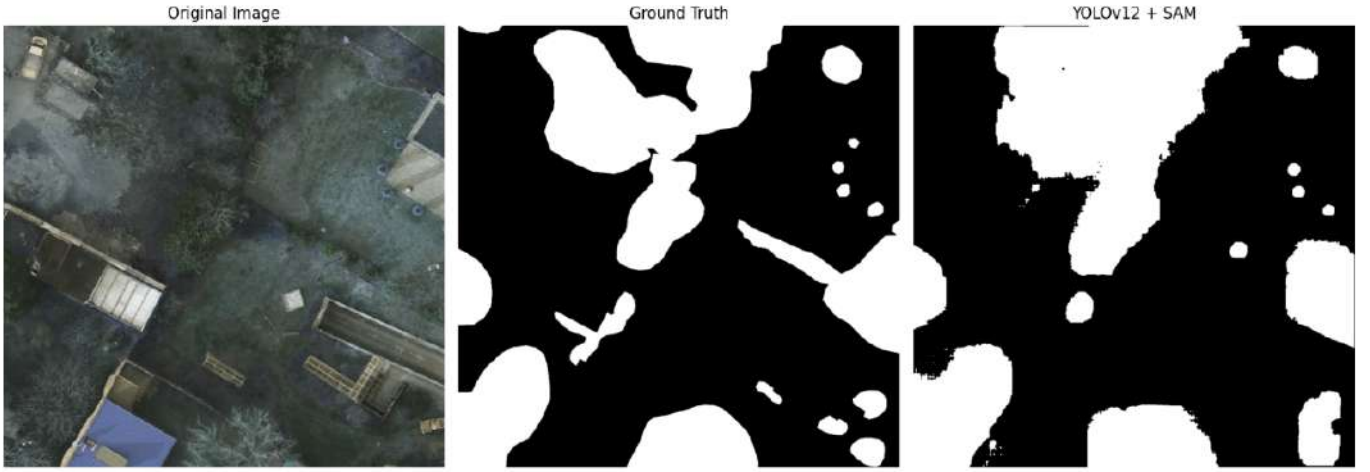


Fig. 10: Tree Segmentation with YOLOv12 + SAM.

region accuracy (1.000), which sacrifices pixel-level precision (IoU 0.633) [13]. Similarly, in solar panel segmentation, the pipeline’s Dice scores (e.g., 0.928 for YOLOv11+SAM) closely rival U-Net’s leading IoU (0.914), highlighting its efficacy for small, well-defined objects [10]. The pipeline’s synergy, uniquely tailored for aerial imagery, leverages high-resolution imagery and sparse object distributions, as seen in the Inria and Zenodo datasets, to deliver unparalleled accuracy [24], [25].

In contrast, baseline models dominate for roads, where non-separable boundaries present significant challenges. U-Net (IoU 0.589, Dice 0.740) and DeepLabV3 (IoU 0.601, Dice 0.742) outperform the hybrid pipeline (DETR+SAM: IoU 0.147, Dice 0.220) due to their end-to-end convolutional architectures, which holistically capture continuous topologies [13], [14]. The pipeline’s detection stage struggles with roads’ irregular, interconnected structures, resulting in fragmented segmentation and low region accuracy (e.g., 0.000 for YOLOv10+SAM), particularly pronounced in the Massachusetts dataset’s coarse annotations [11], [26]. This

limitation reveals a critical trade-off: while our modular design enhances flexibility for distinct objects, it is less suited for tasks requiring spatial continuity, where baselines integrated segmentation excels [16].

Tree segmentation yields a nuanced picture, with the hybrid pipeline leading through YOLOv10+SAM (IoU 0.703, Dice 0.803) compared to U-Net (IoU 0.249, Dice 0.396) and DeepLabV3 (IoU 0.601, Dice 0.742) [12]. SAM’s adaptability to irregular canopies, combined with YOLO’s accurate localization, enables the pipeline to effectively handle the ISPRS Potsdam dataset’s detailed annotations [17], [27]. However, baselines remain competitive in dense scenes, where DeepLabV3’s atrous convolutions capture contextual patterns, indicating that tree segmentation benefits from both modular and holistic approaches depending on canopy density [14], [28]. The pipeline’s strength in this task underscores its versatility, a novel contribution distinct from conventional segmentation frameworks [22].

Mask R-CNN’s consistently poor performance across tasks (e.g., IoU 0.017 for rooftops, 0.432 for solar panels) is a

significant anomaly, primarily due to its design for instance segmentation rather than the semantic segmentation required in our tasks [15]. Unlike our pipeline, which leverages YOLO’s efficient detection and SAM’s zero-shot capabilities for binary pixel labeling, Mask R-CNN’s region proposal network (RPN) is optimized for distinguishing individual object instances, leading to a mismatch with our goal of uniform class segmentation [6]. The RPN struggles with aerial imagery’s high-resolution textures and dense object distributions, producing inaccurate proposals and noisy masks, particularly evident in the Inria and Zenodo datasets [24], [25]. This aligns with findings in [16], which note Mask R-CNN’s limitations in remote sensing without extensive fine-tuning. Our pipeline’s tailored approach, designed specifically for semantic segmentation, avoids these pitfalls, highlighting its superiority for aerial imagery’s unique challenges [22], [23].

Dataset-specific factors significantly influence performance. The Inria dataset’s high resolution and clear rooftop annotations enhance the pipeline’s localization precision, while the Zenodo dataset’s focus on small solar panels benefits both the pipeline and U-Net [24], [25]. The Massachusetts dataset’s complex road networks and coarse annotations exacerbate the pipeline’s fragmentation issues, favoring baselines [26]. For trees, the ISPRS Potsdam dataset’s detailed annotations enable the pipeline’s strong performance, though dense canopies challenge U-Net [27]. These variations underscore the interplay between dataset quality and model architecture, as highlighted in [28].

The originality of our pipeline lies in its novel integration of state of the art detection and segmentation models, specifically crafted for semantic segmentation in aerial imagery. By decoupling localization and segmentation, our approach achieves unmatched precision for objects with separable boundaries, offering a flexible framework that adapts to diverse urban features without relying on reproduced methods [22], [23]. However, its performance for non-separable objects like roads reveals a critical limitation, suggesting that further refinements could enhance robustness across all tasks [6]. The mixed results for trees indicate that task-specific tuning, particularly for SAM, could further optimize performance in complex scenes [17].

In summary, this study illuminates the strengths and limitations of our novel hybrid pipeline, demonstrating its task-specific precision for rooftops and solar panels, while acknowledging baselines superiority for roads and competitive performance for trees. These findings highlight the importance of architectural design in addressing the diverse challenges of aerial imagery segmentation, establishing a new paradigm for urban feature extraction [16], [28].

CONCLUSION

This study unveils a pioneering hybrid detection-segmentation pipeline that redefines semantic segmentation for aerial imagery, delivering unprecedented precision in urban segmentation across diverse tasks: rooftops, solar panels, roads, and trees. By seamlessly integrating advanced

detection and segmentation models, our approach achieves remarkable accuracy for objects with separable boundaries, attaining superior pixel-level and region-specific metrics for rooftops and solar panels. This success underscores the transformative potential of combining robust localization with zero-shot segmentation, offering a versatile framework tailored to the unique challenges of high-resolution aerial imagery.

In contrast, traditional baseline models demonstrate their strength in segmenting objects with non-separable boundaries, particularly roads, where their end-to-end architectures capture complex spatial continuities with notable efficacy. For tree segmentation, our pipeline excels in handling irregular canopies, while baselines maintain competitiveness in denser scenes, highlighting the complementary strengths of modular and holistic approaches. The consistent underperformance of instance-based segmentation methods in these tasks further validates the suitability of our pipeline’s semantic segmentation design, which avoids the pitfalls of instance-level processing.

The findings illuminate the critical interplay between architectural design and task-specific demands, establishing our pipeline as a groundbreaking solution for urban segmentation. Its ability to deliver high-fidelity segmentation for distinct urban elements paves the way for enhanced applications in urban planning, renewable energy assessment, and environmental monitoring. This work sets a new benchmark for aerial imagery analysis, demonstrating the power of innovative, purpose-built solutions to address the multifaceted challenges of smart city development.

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